



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 05

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Program:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int A[2][2], B[2][2], C[2][2];
```

```
    int i, j, k;
```

```
    printf("Enter matrix A:\n");
```

```
    for (i = 0; i < 2; i++)
```

```
        for (j = 0; j < 2; j++)
```

```
            scanf("%d", &A[i][j]);
```

```
    printf("Enter matrix B:\n");
```

```
    for (i = 0; i < 2; i++)
```

```
        for (j = 0; j < 2; j++)
```

```
            scanf("%d", &B[i][j]);
```

```
    for (i = 0; i < 2; i++)
```

```
        for (j = 0; j < 2; j++)
```

```

    {
        C[i][j] = 0;
        for (k = 0; k < 2; k++)
            C[i][j] += A[i][k] * B[k][j];
    }

    printf("Result:\n");
    printf("[[%d,%d],[%d,%d]]",
        C[0][0], C[0][1], C[1][0], C[1][1]);

    return 0;
}

```

Output:

main.c	Run ▶	Output
<pre> 1 #include <stdio.h> 2 3 int main() 4 { 5 int A[2][2] = {{1,2},{3,4}}; 6 int B[2][2] = {{5,6},{7,8}}; 7 int C[2][2] = {0}; 8 int i, j, k; 9 10 for(i = 0; i < 2; i++) 11 for(j = 0; j < 2; j++) 12 for(k = 0; k < 2; k++) 13 C[i][j] += A[i][k] * B[k][j]; 14 15 printf("[[%d,%d],[%d,%d]]", 16 C[0][0], C[0][1], C[1][0], C[1][1]); 17 18 return 0; 19 }</pre>		<pre> [[19,22],[43,50]]</pre>